

Scalar and Vector Quantization for Learned Image Compression: A Study on the Effects of MSE and GAN Loss in Various Spaces

Jonas Löhdefink¹ Fabian Hüger² Peter Schlicht² Tim Fingscheidt¹

Abstract—Recently, learned image compression by means of neural networks has experienced a performance boost by the use of adversarial loss functions. Typically, a generative adversarial network (GAN) is designed with the generator being an autoencoder with quantizer in the bottleneck for compression and reconstruction. It is well known from rate-distortion theory that vector quantizers provide lower quantization errors than scalar quantizers at the same bitrate. Still, learned image compression approaches often use scalar quantization instead. In this work we provide insights into the image reconstruction quality of the often-employed uniform scalar quantizers, non-uniform scalar quantizers, and the rarely employed but bitrate-efficient vector quantizers, all being integrated into backpropagation and operating under the exact same bitrate. Further interesting insights are obtained by our investigation of an MSE loss and a GAN loss. We show that vector quantization is always beneficial for the compression performance both in the latent space and the reconstructed image space. However, image samples demonstrate that the GAN loss produces the more pleasing reconstructed images, while the non-adversarial MSE loss provides better quality scores of various instrumental measures both in the latent space and on the reconstructed images.

I. INTRODUCTION

The non-differentiability of quantizer characteristics poses an inherent problem for neural network learning and so far, learned image compression works such as [1] and [2] use simple scalar quantization approaches with respective quantizer approximations. We see large potential in vector quantization for either lower bitrates or better quality of future learned image codecs, since it is well known from rate-distortion theory [3] that vector quantization always provides a higher coding efficiency compared to scalar quantization which is a first major motivation of this work.

Moreover, generative adversarial networks (GANs) achieve striking results in both image generation and reconstruction, the latter playing the role of an image decoder [1], [4]. Incorporating adversarial loss functions into learned image compression by self-supervised autoencoders constituting the generator of a GAN, see Fig. 1, guarantees to generate realistic outputs even at very low bitrates and has benefits, e.g., in real-time applications dealing with data-intensive machine learning functions. The autoencoder consisting of an encoder, quantizer in the bottleneck,

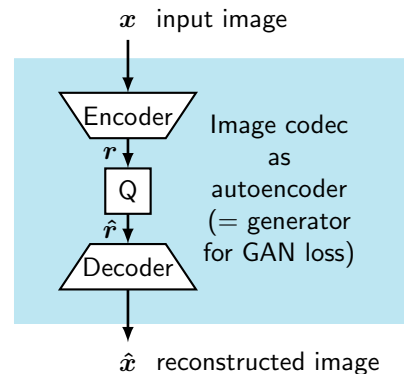


Fig. 1: Autoencoder architecture for learned image compression with an integrated quantizer (uniform or non-uniform scalar quantizer or vector quantizer).

and a decoder, performs compression both w.r.t. image resolution (dimensionality reduction) and w.r.t. value representation (quantization). Learned compression implies an implementation by neural networks [1], [2], [4]–[8], while for transmission, the data is encoded on the transmitter side and quantized before being decoded analogously on the receiver side [2], [4], [9], [10].

Combining an adversarial loss function with a vector quantizer, a second goal of this work is to investigate how the GAN loss influences the performance of various quantization techniques, one of these being vector quantization.

In this paper, we train a neural network for image compression with integrated (vector) quantization without (MSE loss only) and with an adversarial loss function constructing a GAN. We show how to train with integrated quantization, via quantizer-less pretraining, codebook generation, and quantization finetuning. As a third important contribution we provide an evaluation synopsis on the Cityscapes dataset [11], employing several well-known quality measures both in the reconstructed image space, but also in the latent space.

The rest of this paper is organized as follows. Section II gives an overview of prior works. After introducing our framework in Section III, the main quantization options are laid out in Section IV. In Section V, we present our experimental setup and discuss the results. Finally, in Section VI, we provide conclusions.

II. RELATED WORK

The following sections present an overview of recent works on learned image compression, generative adversarial networks, and quantization approaches.

Learned Image Compression: Conventional image codecs such as JPEG [12] and JPEG2000 [13] will soon

¹Jonas Löhdefink, and Tim Fingscheidt are with the Institute for Communications Technology, Technische Universität Braunschweig, Schleinitzstr. 22, 38106 Braunschweig, Germany {j.loehdefink, t.fingscheidt}@tu-bs.de

²Fabian Hüger, and Peter Schlicht are with Volkswagen Group Innovation, Berliner Ring 2, 38440 Wolfsburg, Germany {fabian.hueger, peter.schlicht}@volkswagen.de